
Polynomial Links as Parameter-Tied Rank Lifts for Tucker Tensors

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Abstract

1 Tucker decomposition is a compact model for multiway data, but its observation
2 map is linear in the reconstructed latent tensor. Nonlinear sensing and imaging
3 responses can therefore make a low-rank latent tensor appear higher-rank in the
4 observed domain. We propose Nonlinear Tucker Decomposition with Polynomial
5 Links (NTD-PL), a Tucker enhancement that learns a shared entrywise polynomial
6 observation link jointly with a Tucker backbone. NTD-PL keeps the backbone
7 ranks fixed and adds only scalar link coefficients. Its polynomial powers induce a
8 parameter-tied rank lift: each degree forms Veronese-lifted Tucker factors tied to
9 the same latent tensor, rather than an untied high-rank correction.

10 We analyze this structure through Veronese rank expansion, separation from
11 matched-rank Tucker, and approximation results for shared nonlinear observa-
12 tion maps. We optimize a masked reconstruction and completion objective by
13 alternating Link Refresh, a small ridge update for the scalar link, with Linked-
14 Backbone Descent and degree continuation. Experiments on controlled tensors,
15 CAVE hyperspectral reconstruction and completion, rank-lift and Link Refresh
16 diagnostics, and external cross-domain cases show scoped gains: NTD-PL helps
17 when the residual left by matched-rank Tucker is coherent with a shared entry-
18 wise nonlinear response, and is not intended to replace rank selection or general
19 nonlinear tensor decoders.

20 1 Introduction

21 Tensor decompositions provide compact models for multiway data in scientific sensing, imaging,
22 and signal processing [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013]. Tucker
23 decomposition is a standard choice because it separates a small core from mode-wise factors. Its
24 observation map, however, is linear: the reconstructed Tucker entry is fitted directly to the measured
25 entry. This assumption is natural for many data sets, but it can be mismatched to sensing and imaging
26 pipelines with response curves, saturation, calibration effects, or material-dependent nonlinearities
27 [Debevec and Malik, 1997, Grossberg and Nayar, 2004, Bioucas-Dias et al., 2012, Heylen et al.,
28 2014]. In such settings, the latent scene may be well organized by a low-rank Tucker tensor while the
29 measured tensor appears to have larger multilinear rank after a shared entrywise response.

30 A direct remedy is to increase Tucker rank, but this changes the multilinear budget and expands the
31 core and factors throughout the model. More flexible nonlinear tensor decoders based on kernels,
32 Gaussian processes, or neural networks can also be useful, but they address a broader problem and
33 often make the relation to a Tucker backbone less explicit [Xu et al., 2011, Zhe et al., 2016, Liu et al.,
34 2017, Saragadam et al., 2024]. We ask a narrower question: can a Tucker model keep its chosen
35 backbone ranks fixed while learning a shared entrywise response that explains coherent nonlinear
36 residuals left by matched-rank Tucker?

We instantiate this idea with a polynomial link, a classical low-dimensional tool for modeling nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$$

be a Tucker latent tensor. Instead of fitting the observation directly by $\mathcal{S}$, we model

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p},$$

where powers are entrywise. We call the model Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case, keeps the Tucker backbone and multilinear ranks fixed, and adds only $P + 1$ scalar link coefficients. The higher powers $\mathcal{S}^{\odot p}$ induce high-order interactions among the same core and factors, so the added expressivity is parameter-tied to one learned latent tensor.

This viewpoint is central to the paper. NTD-PL is best understood as a Tucker enhancement and shared-backbone rank lift, not as a generic nonlinear decoder. A degree- p power of the latent tensor behaves like an ordinary Tucker tensor whose factors are polynomial feature lifts of the original factors, but those lifted features remain tied to one learned backbone. The new degrees therefore reshape the low-rank geometry through a shared observation link, rather than adding an untied high-rank tensor block. This is also why the link is not a post-hoc calibration layer: the Tucker backbone is learned through the polynomial observation map, and the latent tensor is reshaped jointly with the scalar response, rather than fitting Tucker first and calibrating a fixed output.

Our claims are intentionally scoped. NTD-PL is not meant to replace rank selection, and increasing Tucker rank remains the right comparison when the residual is ordinary multilinear structure. Nor do we claim to dominate neural, kernel, or other nonlinear tensor factorization methods. The intended regime is more specific: the residual left by matched-rank Tucker should be coherent with a shared entrywise nonlinear response.

The paper develops this regime from four angles. Section 3 defines the shared polynomial observation link and its tied high-order Tucker channels. Section 4 proves Veronese rank expansion, a separation from matched-rank Tucker, and approximation results for shared nonlinear observation maps. Section 5 gives the masked objective and an alternating solver with Link Refresh, Linked-Backbone Descent, and degree continuation. Section 6 tests the mechanism on controlled tensors, CAVE hyperspectral reconstruction and completion, rank-lift and Link Refresh diagnostics, and external cross-domain boundary cases. We discuss failure modes and scope in Section 7.

Contributions. We make the following contributions.

- **Model.** We propose NTD-PL, a Tucker enhancement with a shared polynomial observation link on a Tucker latent tensor. It keeps the Tucker backbone and multilinear ranks fixed while adding only scalar coefficients for a parameter-tied, entrywise response.
- **Theory.** We characterize the induced model class through Veronese rank expansion, separation from matched-rank Tucker, and approximation results for shared nonlinear observation maps.
- **Optimization.** We derive a masked objective for reconstruction and completion, and optimize it with Link Refresh, Linked-Backbone Descent, and degree continuation.
- **Experiments.** We evaluate controlled nonlinear residuals, hyperspectral reconstruction and completion, rank-lift and Link Refresh diagnostics, and external cross-domain boundary cases.

2 Related Work

Low-rank tensor models and Tucker backbones. CP, Tucker, tensor train, and tensor ring decompositions provide compact representations for multiway data through different rank notions and factor topologies [Kolda and Bader, 2009, Sidiropoulos et al., 2017, De Lathauwer et al., 2000, Oseledets, 2011, Zhao et al., 2016]. Tucker is the closest baseline for NTD-PL because it uses the same core-factor multilinear backbone. NTD-PL does not primarily compete by changing the rank tuple, replacing the factor topology, or moving to another tensor-network format. Instead, it keeps the Tucker backbone and multilinear ranks fixed and changes the observation map on top of the Tucker latent tensor.

84 **Nonlinear tensor factorization and neural/kernel decoders.** Nonlinear tensor models include
85 generalized likelihood and link formulations, Bayesian and kernelized tensor factorizations, Gaussian-
86 process tensor models, and neural tensor decoders [Hong et al., 2020, Schein et al., 2016, Xu
87 et al., 2011, Zhe et al., 2016, Liu et al., 2017, Pan et al., 2020, Saragadam et al., 2024, Varolgunes
88 et al., 2023]. These methods can be more flexible than a shared scalar polynomial link, but this
89 flexibility often comes with heavier inference, additional hyperparameters, or a less transparent finite-
90 dimensional relation to Tucker. NTD-PL is complementary: it deliberately uses a low-dimensional
91 shared entrywise response rather than a general kernel, Gaussian-process, or neural decoder, so that
92 the induced effect can be analyzed relative to a fixed Tucker backbone.

93 **Polynomial feature lifts and high-order interactions.** Polynomial and Volterra-style expansions
94 are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985].
95 Related matrix and tensor methods use polynomial feature maps, quadratic factorizations, or explicit
96 high-order interactions to enlarge a linear model [Zhai et al., 2024, Liu et al., 2015, Deng et al., 2016].
97 Such constructions can make nonlinear interactions explicit, but untied high-order parameterizations
98 can increase parameter dimension and contraction cost. NTD-PL does not introduce an independent
99 high-order tensor parameter. Its polynomial powers are powers of one learned Tucker latent tensor,
100 inducing parameter-tied high-order interactions and Veronese-style shared-backbone rank lifts.

101 **Positioning.** Existing nonlinear tensor models either use broad nonlinear decoders, modify like-
102 lihoods or kernels, or explicitly parameterize high-order interactions. What is less developed is a
103 Tucker-preserving, low-dimensional, analyzable polynomial observation link that yields a parameter-
104 tied rank lift and supports masked learning. NTD-PL trades general nonlinear flexibility for structure,
105 interpretability, and a direct finite-dimensional rank-lift view. Its expected benefit is therefore scoped
106 to settings where matched-rank Tucker leaves a coherent nonlinear residual explained by a shared
107 entrywise response.

108 3 Model

109 3.1 Model and Observation Assumption

110 Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ be an observed tensor and let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad \mathcal{X} \in \mathbb{R}^{R_1 \times \dots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \quad (1)$$

111 NTD-PL treats $\mathcal{S}$ as a shared low-rank latent tensor and models each observed entry through a scalar
112 polynomial response:

$$Y_{\mathbf{i}} = f_{\beta}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \quad f_{\beta}(s) = \sum_{p=0}^P \beta_p s^p, \quad \mathbf{i} = (i_1, \dots, i_N). \quad (2)$$

113 Equivalently, the tensor prediction is

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}, \quad \beta \in \mathbb{R}^{P+1}. \quad (3)$$

114 The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel,
115 and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is
116 recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the entry-generation map
117 while leaving the Tucker backbone and its multilinear ranks fixed.

118 3.2 Shared-Backbone High-Order Interactions

119 Although f_{β} is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. We arrive at it
120 by first defining an explicit p -fold Tucker extension: instead of mapping one latent index per mode,
121 the mode- n projection is a higher-order tensor $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$ that couples p latent indices. The
122 case $p = 1$ is ordinary Tucker, while $p = 2$ gives a quadratic Tucker-style interaction between two
123 latent Tucker tuples. Appendix A.1 gives the formal entrywise construction and tensor-network view.

124 The explicit construction is a reference model, not the parameterization we train. Its projection
 125 tensors grow as $I_n R_n^p$, and each degree would add a new set of mode projections. NTD-PL keeps the
 126 same high-order interaction pattern by tying the p -fold projection to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}.$$

127 Under this tying, the degree- p channel is exactly S_i^p . Thus it still couples p latent Tucker index tuples,
 128 but all copies share the same mode factors. Across degrees, the only new free parameters are the
 129 scalar coefficients $\{\beta_p\}$, which mix these tied channels.

130 The latent tensor $\mathcal{S}$ and the coefficients β are fitted jointly, so the low-rank backbone is optimized
 131 together with the polynomial channels. Section 4 analyzes the induced rank expansion and the
 132 resulting separation from standard Tucker.

133 3.3 Masked Least-Squares Objective

134 Let $\Omega \in \{0, 1\}^{I_1 \times \dots \times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_i \Omega_i$. We estimate the core,
 135 factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^N \frac{\lambda_n}{2} \|\mathbf{A}^{(n)}\|_F^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2. \quad (4)$$

136 The same formulation covers full reconstruction by taking $\Omega \equiv \mathbf{1}$ and tensor completion by restricting
 137 the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the
 138 low-dimensional polynomial link.

139 4 Theory

140 The theory addresses the rank-lift view raised by the model definition. A scalar link has few free
 141 parameters, but its powers can induce ordinary Tucker tensors with larger multilinear ranks. We
 142 first make this induced structure explicit, then give a separation from matched-rank Tucker and an
 143 approximation route for shared nonlinear observation maps.

144 4.1 Rank Expansion and Separation

145 **Proposition 1** (Shared-backbone rank expansion). *Let $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ have Tucker rank*
 146 *at most $(R_1, \dots, R_N)$. For every integer $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks*
 147 *at most*

$$\left(\binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \quad (5)$$

148 *More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original*
 149 *factors. For $\mathbf{A} \in \mathbb{R}^{I \times R}$, define $\mathbf{A}^{(p)} \in \mathbb{R}^{I \times D}$, $D = \binom{R+p-1}{p}$, by*

$$A_{i, \alpha}^{(p)} = \prod_{r=1}^R A_{i, r}^{\alpha_r}, \quad (6)$$

150 *where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)(p)}, \dots, \mathbf{A}^{(N)(p)}$.*

151 This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree- p
 152 channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same
 153 backbone factors. The link therefore changes rank structure through tied features, not through an
 154 independent high-order parameter block.

155 **Theorem 1** (Tensorial separation from standard Tucker). *Let $N \geq 3$, $R \geq 2$, $p \geq 2$, and suppose*
 156 *$I_n \geq D$ for every n , where $D = \binom{R+p-1}{p}$. Then there exists an order- N tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ that*
 157 *is representable by a degree- p NTD-PL model with Tucker backbone rank $(R, \dots, R)$, but whose*
 158 *exact standard Tucker rank is*

$$(D, \dots, D). \quad (7)$$

159 Equivalently, a degree- p polynomial link can generate tensors whose multilinear ranks are
 160 $((\binom{R+p-1}{p}, \dots, \binom{R+p-1}{p}))$ from a shared Tucker backbone of rank $(R, \dots, R)$.

161 The proof in Appendix B uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature
 162 matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker
 163 tensors: it identifies a structured high-rank subset generated by a low-rank shared backbone.

164 4.2 Approximation Under Nonlinear Observations

165 **Theorem 2** (Polynomial generation). Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most
 166 $(R_1, \dots, R_N)$ and g is a polynomial of degree at most P . Then degree- P NTD-PL with the same
 167 Tucker rank can represent $\mathcal{Y}$ exactly.

168 **Theorem 3** (Analytic link approximation). Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most
 169 $(R_1, \dots, R_N)$ and $\|\mathcal{S}^*\|_\infty \leq B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and
 170 $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree- P NTD-PL model with the same Tucker backbone
 171 satisfying

$$\frac{1}{\sqrt{M}} \|\mathcal{Y} - \hat{\mathcal{Y}}\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \quad M = \prod_{n=1}^N I_n. \quad (8)$$

172 Thus the same finite link that induces a tied rank lift also gives a direct approximation route for smooth
 173 entrywise responses on a bounded latent range. The statement uses the simple Taylor truncation
 174 bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

175 Together, the results clarify the scope of NTD-PL. Polynomial links can approximate shared nonlinear
 176 observation maps while keeping the same latent Tucker rank, and their powers induce Veronese-
 177 lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without
 178 introducing untied high-order tensor parameters.

179 5 Optimization

180 We optimize the masked objective in Section 3.3 by alternating between the scalar link and the Tucker
 181 backbone. The nonlinear part is low-dimensional once the latent Tucker tensor is fixed, while the
 182 backbone update remains a standard differentiable Tucker fitting step through the current link. We
 183 call the two stages *Link Refresh* and *Linked-Backbone Descent*.

184 **Link Refresh.** Fix the current latent tensor

$$S = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket. \quad (9)$$

185 For active degree $Q \leq P$, updating the polynomial coefficients reduces to a small masked ridge
 186 regression,

$$\min_{\beta_{0:Q}} \frac{1}{2|\Omega|} \|y_\Omega - \Phi_{\Omega,Q}(S) \beta_{0:Q}\|_2^2 + \frac{\lambda_\beta}{2} \|\beta_{0:Q}\|_2^2, \quad (10)$$

187 where $\Phi_{\Omega,Q}(S)$ contains the observed powers $[1, s, \dots, s^Q]$. This step is cheap because it solves
 188 only a $(Q+1) \times (Q+1)$ linear system. It also gives an interpretable view of the link: once the
 189 backbone exposes a one-dimensional latent value at each entry, the refresh estimates the best shared
 190 scalar response on the observed entries.

191 **Linked-Backbone Descent.** With β fixed, the Tucker parameters are updated through the current
 192 nonlinear observation map. The masked residual is first propagated to the latent tensor as

$$T = \frac{1}{|\Omega|} \Omega \odot (f_\beta(S) - Y) \odot f'_\beta(S), \quad (11)$$

193 and then backpropagated through the Tucker reconstruction to update $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$. In implemen-
 194 tation we use Adam for these first-order steps, but the update itself is just standard Tucker reverse
 195 contractions composed with the current link. The backbone is therefore not calibrated after fitting; it
 196 is jointly reshaped under the shared nonlinear link.

197 **Stabilization.** Directly activating all polynomial degrees from the start is brittle, because higher
 198 powers of S amplify scale ambiguity. We therefore begin from the Tucker-compatible degree-one
 199 model and activate higher degrees gradually. When a new degree is switched on, its coefficient is
 200 initialized at zero, so the model expands from the current lower-degree solution without changing
 201 the prediction abruptly. We also normalize Tucker factor scales and absorb the corresponding
 202 scales into the core, which fixes a Tucker gauge without changing S . For the coefficient solve, the
 203 implementation may regress on a centered and scaled latent variable and then map the solution back
 204 to the original power basis.

205 **Cost.** The dominant cost is the same Tucker forward and reverse contractions used by standard
 206 Tucker optimization at the chosen rank. The polynomial link adds only $O(P|\Omega|)$ elementwise work
 207 for evaluating $f_\beta(S)$ and $f'_\beta(S)$, plus a small $(P+1) \times (P+1)$ ridge solve. Since the experiments
 208 use small P , the overhead is modest relative to the Tucker contractions. Full gradients, stabilization
 209 details, and pseudocode are given in Appendix A.2.

210 **Masked-entry uniform convergence.** The masked objective also admits a standard sampling
 211 interpretation. Let $m = |\Omega|$ and let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set, where $d =$
 212 $\prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P+1)$. If the entrywise losses are bounded by $0 \leq \ell_i(\theta) \leq B_\ell$ and
 213 G -Lipschitz in θ , then with probability at least $1 - \delta$,

$$\sup_{\theta \in \Theta} |\mathcal{L}(\theta) - \hat{\mathcal{L}}_m(\theta)| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}, \quad (12)$$

214 for every $0 < \varepsilon \leq B_\Theta$. This is a standard uniform convergence bound, not a sharp minimax result,
 215 but it clarifies that masked training controls the full-entry risk with a complexity term determined by
 216 the tied NTD-PL parameter count. The proof is given in Appendix B.4.

217 6 Experiments

218 This section asks three questions: (i) Does NTD-PL behave as expected under controlled nonlinear
 219 residuals? (ii) Does the polynomial link enhance a Tucker backbone in reconstruction and held-out
 220 completion? (iii) Can we diagnose when the enhancement appears across datasets? Unless stated
 221 otherwise, real tensors are globally max-normalized and used without spatial downsampling or
 222 cropping. RMSE and SAM are used for reconstruction; RMSE* and SAM* are computed only on
 223 held-out missing entries for completion. NTD-PL is compared mainly against Tucker at the same
 224 multilinear rank, except in the explicit rank-lift ablation. We also use a Link Refresh diagnostic score,
 225 D_{link} : after fitting Tucker, run the cheap scalar Link Refresh ridge solve once on the Tucker latent
 226 tensor, and measure the residual reduction on task-specific evaluation entries. Larger D_{link} means the
 227 Tucker residual is more compatible with a shared nonlinear link.

228 6.1 Controlled Nonlinear Residuals

229 This section isolates the mechanism on synthetic tensors. We generate observations as

$$Y_\alpha = \sqrt{1 - \alpha} S^* + \sqrt{\alpha} \tilde{R}^*,$$

230 where S^* is a low-rank Tucker tensor and $\tilde{R}^*$ is the normalized component of a scalar nonlinear
 231 response orthogonal to S^* . Thus α controls nonlinear residual energy rather than the raw amplitude
 232 of a scalar nonlinearity. When the source is linear or nearly linear, NTD-PL should collapse toward
 233 Tucker-like behavior; as nonlinear residual energy increases, it should separate from linear tensor
 234 baselines. The alpha sweep in Figure 1 shows this transition, while the degree-gain heatmap
 235 in Figure 2 ties the gain to the expressive order of the polynomial link. Degree continuation and
 236 coefficient-contribution details are deferred to Appendix C.1, which also reports the linear consistency
 237 negative control.

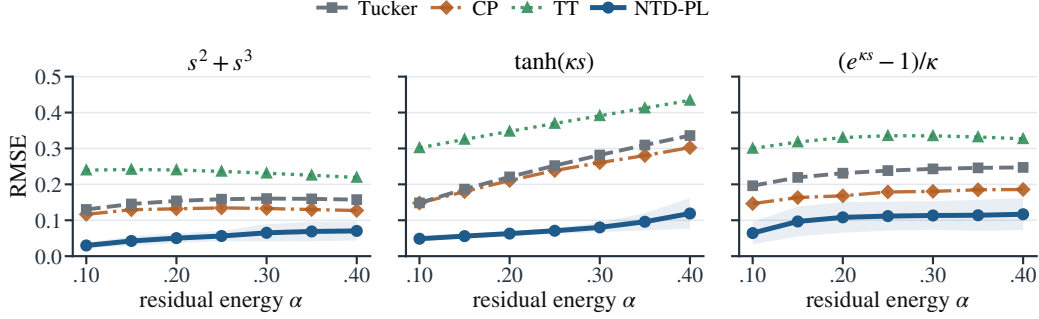


Figure 1: Controlled nonlinear alpha sweep. Each panel uses a different scalar response to define the orthogonal residual; α sets nonlinear residual energy, and lower RMSE is better.

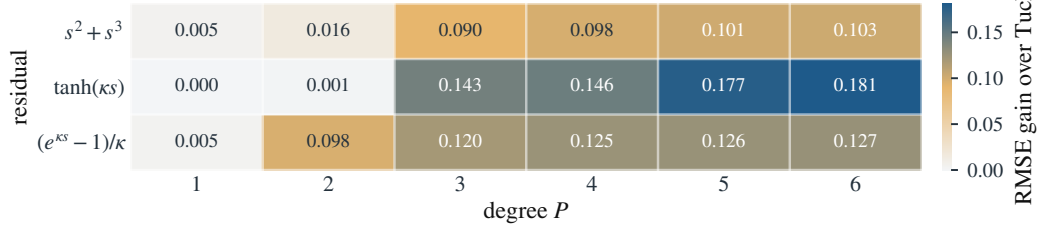


Figure 2: Controlled nonlinear degree-gain heatmap at $\alpha = 0.25$. Cell values are Tucker minus NTD-PL RMSE, so larger values favor NTD-PL.

6.2 Tucker Enhancement on Real Hyperspectral Tensors

This section asks whether NTD-PL enhances a Tucker backbone on real hyperspectral tensors, both when all entries are observed and when evaluation is performed on held-out missing entries.

Representation enhancement. Full-observation reconstruction is a representation-capacity check, not a generalization test. Figure 3 shows that NTD-PL improves both RMSE and SAM on CAFE at matched Tucker backbone rank and similar parameter budget. The advantage shrinks as Tucker rank increases, which is expected: ordinary multilinear capacity already explains more of the tensor. The rank-lifted Tucker ablation in Table 1 clarifies the mechanism further. Tucker can buy back part of the RMSE gap by spending more rank, so the gain is not explained by a small ordinary rank increase alone. They support the view that NTD-PL improves a tight Tucker backbone through a jointly fitted shared link. The joint link-family smoke test in Appendix C.4 further suggests that the effect is tied to joint scalar-link fitting rather than to the polynomial basis alone.

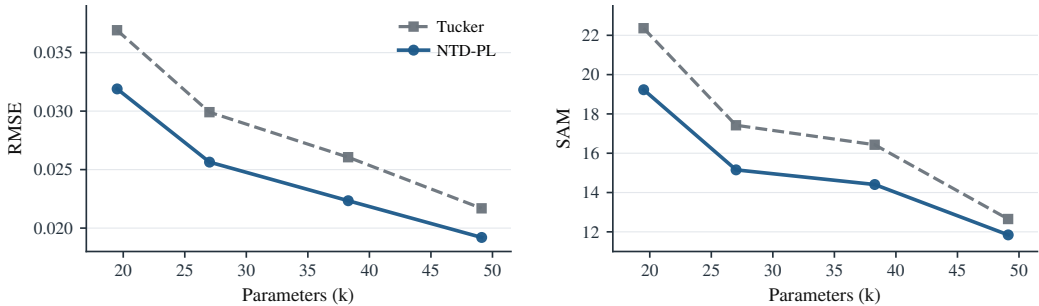


Figure 3: CAFE full-observation reconstruction as a representation-capacity check. Each point averages 15 scenes at a matched Tucker backbone rank. The plotted backbone ranks are (18, 18, 3), (24, 24, 4), (33, 33, 4), and (40, 40, 5). Unlike completion, this measures representational error when all entries are observed.

Held-out enhancement. Masked completion is the stronger test because it asks whether the learned low-rank geometry transfers to unobserved entries. Table 2 reports CAFE completion at several random mask rates, with RMSE* and SAM* computed only on missing entries. Under the same rank

Table 1: Rank-lifted Tucker comparison on CAVE reconstruction. For each NTD-PL backbone rank, we increase Tucker’s spatial ranks while keeping the spectral rank fixed, and report the first Tucker rank that matches the NTD-PL RMSE.

NTD-PL reference				First Tucker matching RMSE				Extra params
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
(18, 18, 3)	20k	0.0319	19.23	(28, 28, 3)	31k	0.0316	21.41	+59.5%
(24, 24, 4)	27k	0.0256	15.15	(35, 35, 4)	41k	0.0254	16.26	+51.3%
(33, 33, 4)	38k	0.0223	14.41	(49, 49, 4)	60k	0.0222	15.28	+56.5%

budget as Tucker, NTD-PL gives consistent held-out RMSE* gains and usually improves SAM* as well. The paired sign and Wilcoxon tests in Appendix C.2 show the same pattern scene by scene.

Table 2: Full-resolution CAVE completion at rank (33, 33, 4) under random masks. Here ρ denotes the missing-entry rate. RMSE* and SAM* are computed on missing entries. Tucker and NTD-PL columns report mean $\pm$ standard deviation over scenes; gains are paired relative improvements over Tucker.

ρ	RMSE*↓			SAM*↓		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263 $\pm$ 0.0179	0.0229$\pm$0.0158	12.9%	11.82 $\pm$ 5.33	8.36$\pm$3.38	29.3%
0.3	0.0263 $\pm$ 0.0179	0.0230$\pm$0.0158	12.5%	14.75 $\pm$ 6.33	12.47$\pm$5.06	15.5%
0.5	0.0264 $\pm$ 0.0179	0.0230$\pm$0.0158	12.9%	15.56 $\pm$ 6.85	13.48$\pm$5.75	13.4%
0.7	0.0266 $\pm$ 0.0180	0.0219$\pm$0.0128	17.7%	16.02 $\pm$ 7.21	13.79$\pm$5.43	13.9%

6.3 Diagnosing Nonlinear Low-Rank Structure Across Data

NTD-PL is not expected to help uniformly on all tensors. Its gain should appear when a matched-rank Tucker fit leaves a coherent residual that a shared scalar link can reduce. We denote this diagnostic by D_{link} : after fitting Tucker, we freeze the Tucker backbone, run one scalar Link Refresh ridge solve, and measure the residual-energy reduction on the relevant evaluation entries. Larger D_{link} indicates a stronger shared entry-level nonlinear residual:

$$D_{\text{link}} = 10 \log_{10} \frac{\|Y_{\text{eval}} - \hat{Y}_T\|_F^2}{\|Y_{\text{eval}} - \hat{Y}_{\text{link}}\|_F^2},$$

where $\hat{Y}_T$ is the Tucker output and $\hat{Y}_{\text{link}}$ is the output after the one-step refresh.

For HSI, we use natural scene-level tensor units rather than artificial patches. CAVE already provides multiple scene tensors, so Figure 4 evaluates the scene-level relationship between D_{link} and NTD-PL held-out RMSE* gain.

Figure 4 first validates the diagnostic on CAVE completion: scenes with higher D_{link} receive larger held-out RMSE* gains. We then extend the same protocol to non-HSI units. For natural images, each image is treated as one tensor unit; for object-view datasets, each object or instance forms a multi-view tensor unit; for action video, each KTH segment or UCF101 clip is treated as one action-clip tensor unit. We do not concatenate unrelated images, objects, or clips into a giant tensor, since D_{link} tests whether a single shared scalar link is coherent within one unit.

Table 3 summarizes the non-HSI diagnosis on full-observation reconstruction units. Natural images show weaker shared-link yield, especially CIFAR-10, whose median D_{link} is close to zero and whose low-yield tercile has almost no gain. Object-view tensors show clearer effective units: COIL-100 and smallNORB have larger median D_{link} , stronger rank-matched gains, and high-yield terciles with larger improvements. Action-video clips show moderate shared-link yield: KTH-Action is stronger than UCF101, but both datasets retain high diagnostic correlation and visibly larger gains in their high-yield terciles than in their low-yield terciles. These results support the scoped claim that NTD-PL helps when the Tucker residual contains shared link-like structure, rather than uniformly across all visual tensors. Representative units selected only by D_{link} are reported in Appendix C.7: high- D_{link} object-view units such as COIL object 91 and smallNORB category 2 instance 7 show large gains, while low- D_{link} boundary units show small or negative gains.

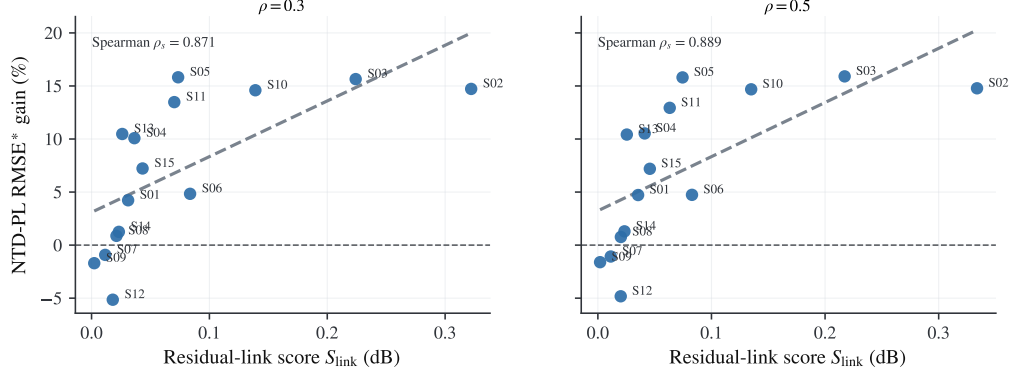


Figure 4: CAVE completion at $\rho \in \{0.3, 0.5\}$: scene-level NTD-PL RMSE* gain versus Link Refresh diagnostic D_{link} . The left and right panels use $\rho = 0.3$ and $\rho = 0.5$, respectively, and each panel annotates its Spearman coefficient.

Table 3: Cross-domain Link Refresh diagnosis. Each unit is independently fitted by Tucker and NTD-PL at matched backbone rank. D_{link} is computed by one frozen-backbone Link Refresh step and diagnostic tiers are assigned from D_{link} , not from NTD-PL gain. Natural images use per-image units; object-view datasets use per-object or per-instance view tensors; action-video datasets use per-clip units.

Domain	Dataset	Unit	#Units	Med. D_{link}	Spearman	Med. gain	Low/High gain
Natural images	CBSD68	image	16	0.075	0.835	5.77%	3.87/12.44%
Natural images	CIFAR-10	image	1000	0.008	0.934	2.35%	-0.01/7.85%
Object-view	COIL-100	object-view tensor	100	0.208	0.932	5.40%	2.77/9.14%
Object-view	smallNORB	instance-view tensor	25	0.264	0.945	9.84%	6.92/14.16%
Action video	KTH-Action	action clip	240	0.107	0.884	4.32%	2.74/10.87%
Action video	UCF101	action clip	240	0.065	0.890	2.89%	1.44/5.89%

7 Limitations

NTD-PL assumes that a common entrywise nonlinear response explains a coherent part of the residual left by matched-rank Tucker. This shared scalar link may be less effective when nonlinearities vary strongly across spatial regions, spectral bands, modes, or samples. In such cases, a single polynomial response can underfit the domain variation even if the Tucker backbone is appropriate.

The method also does not remove rank selection. NTD-PL enhances a fixed Tucker rank budget by changing the observation map on top of the latent tensor; it should not be interpreted as eliminating the need for larger ranks when the residual is ordinary multilinear structure. Increasing Tucker rank remains an important baseline and model-selection choice.

Polynomial degree introduces a second tradeoff. Higher degrees increase the approximation capacity of the shared link, but they can amplify Tucker scale ambiguity and numerical conditioning issues in the power basis. This motivates the ridge-regularized Link Refresh step, factor normalization, and degree continuation used in the optimizer.

Finally, the joint problem remains nonconvex. The Link Refresh subproblem is a small ridge regression once the latent tensor is fixed, but the Tucker backbone and polynomial link are learned jointly through a nonconvex objective. We do not claim global optimality.

Our strongest empirical evidence is currently from controlled nonlinear residuals and hyperspectral reconstruction and completion. Broader validation on large sparse tensors, temporal or multimodal tensors, and domain-varying nonlinear responses is left for future work.

8 Conclusion

This paper develops polynomial links as parameter-tied rank lifts for Tucker tensors. The model keeps a Tucker latent tensor as the backbone and learns a shared entrywise polynomial observation link,

304 adding only scalar coefficients. The resulting powers of the same latent tensor act as shared-backbone
 305 high-order interactions rather than untied high-rank corrections.

306 The analysis makes this rank-lift view explicit through Veronese rank expansion, separation from
 307 matched-rank Tucker, and approximation results for shared nonlinear links. The optimization uses a
 308 masked objective with Link Refresh, Linked-Backbone Descent, and degree continuation, while the
 309 experiments and diagnostics identify the intended regime: the link helps when matched-rank Tucker
 310 leaves a coherent nonlinear residual explained by a shared entrywise response.

311 Future work may consider spatially or spectrally adaptive links, scalable sparse implementations, and
 312 richer structured links beyond scalar polynomials.

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376 A Additional Model Details

377 A.1 Explicit p -Fold Tucker Motivation

378 This section expands the motivation behind the explicit p -fold Tucker channel in Section 3.2. The
379 construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map.
380 In the matrix case, one may move from a linear model $\mathbf{y} = \mathbf{A}\mathbf{x}$ to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

381 where the second term couples pairs of latent coordinates. The corresponding question for Tucker is:
382 if standard Tucker is the multilinear low-rank map for multiway data, what is the direct p -th order
383 interaction analogue?

384 For a Tucker backbone, an explicit p -fold interaction can be written as

$$\tilde{Y}_i^{(p)} = \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)}. \quad (13)$$

385 Here each mode projection depends on p latent indices instead of one. For $p = 1$, this reduces to the
386 ordinary Tucker entry formula. For $p = 2$, the same core appears twice and the mode projections
387 couple two latent tuples, giving a quadratic Tucker-style channel. Figure 5 illustrates this progression.

388 The explicit construction is useful as a reference model, but it is not the parameterization used by
389 NTD-PL. Its mode projections $\mathcal{A}^{(n,p)}$ have size $I_n R_n^p$, so a direct implementation introduces new

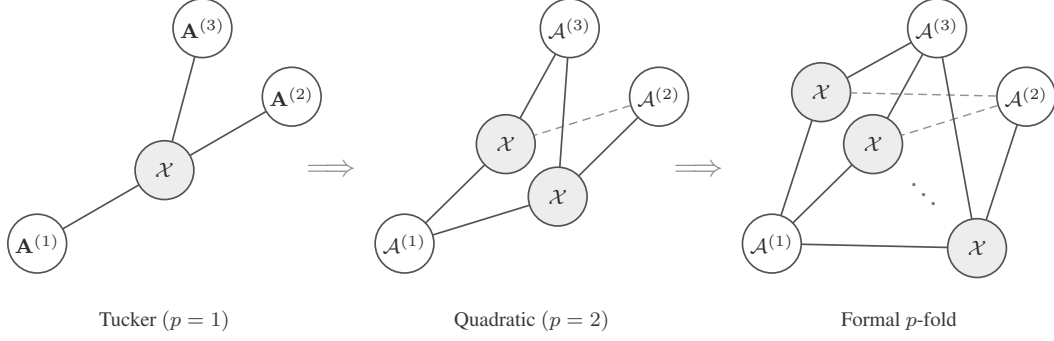


Figure 5: Tensor-network view of a formal p -fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} \quad (14)$$

Substituting (14) into (13) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$. Thus the degree- p NTD-PL channel is a shared-projection version of the explicit p -fold Tucker prototype.

This is the modeling role of the p -fold Tucker prototype: it defines the high-order Tucker interaction that NTD-PL compresses. The final model does not train the untied tensors $\mathcal{A}^{(n,p)}$; it keeps a single Tucker backbone and adds only the polynomial coefficients $\{\beta_p\}$.

A.2 Optimization Details

This section gives the update equations behind the solver summarized in Section 5. The implementation follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression updates of the link coefficients.

Backbone gradients. For a fixed β , define

$$f'_\beta(\mathcal{S}) = \sum_{p=1}^P p \beta_p \mathcal{S}^{\odot(p-1)}. \quad (15)$$

The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - \mathcal{Y}) \odot f'_\beta(\mathcal{S}). \quad (16)$$

Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \dots \times_N \mathbf{A}^{(N)\top} + \lambda_X \mathcal{X}. \quad (17)$$

For the n -th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \dots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \dots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \quad (18)$$

Then

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \quad (19)$$

The experiments use Adam for these first-order steps. This choice is not part of the model definition; any comparable first-order optimizer can use the same gradients.

409 **Link update.** With S fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is the ridge
 410 regression in (10). Its closed-form solution is

$$\beta_{0:Q}^* = (\Phi_{\Omega,Q}^\top \Phi_{\Omega,Q} + |\Omega| \lambda_\beta \mathbf{I})^{-1} \Phi_{\Omega,Q}^\top \mathbf{y}_\Omega. \quad (20)$$

411 Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal
 412 equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In
 413 the code, this small system is solved either from an explicit Vandermonde design or from moment
 414 equations built from cached powers.

415 **Stabilization and continuation.** High powers of S can be sensitive to scale. The implementation
 416 therefore normalizes factor columns after backbone updates and absorbs the corresponding scales
 417 into the core; this is a gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For the
 418 coefficient solve, it may regress on the centered and scaled latent variable $(S - \mu)/\sigma$, then convert
 419 the solution back to the original power basis of S .

420 The active degree follows the schedule

$$t_k = \text{round} \left(\frac{kT}{P} \right), \quad k = 1, \dots, P-1.$$

421 When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new
 422 coefficient β_{p+1} is initialized at zero.

423 **Cost.** The dominant cost is forming the Tucker reconstruction and its reverse contractions on the
 424 observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same
 425 rank, plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms
 426 a Vandermonde design or the corresponding moment equations and solves a $(P+1) \times (P+1)$ linear
 427 system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the
 428 Tucker contractions.

429 Algorithm 1 gives the full procedural summary.

Algorithm 1 Masked NTD-PL optimization

Require: observed tensor $\mathcal{Y}$, mask Ω , rank $(R_1, \dots, R_N)$, degree P

```

1: initialize  $\mathcal{X}, \{\mathbf{A}^{(n)}\}$  from Tucker on the masked tensor
2: initialize active degree  $P_{\text{act}} \leftarrow 1$  and  $\beta = (0, 1)$ 
3: for  $t = 1, \dots, T$  do
4:   form  $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ 
5:   if a link update is scheduled then
6:     update active coefficients  $\beta_{0:P_{\text{act}}}$  by ridge regression on  $\{(S_i, Y_i) : \Omega_i = 1\}$ 
7:   end if
8:   compute  $f_\beta(\mathcal{S})$  and  $f'_\beta(\mathcal{S})$ 
9:   take first-order steps on  $\mathcal{X}$  and  $\{\mathbf{A}^{(n)}\}$  using the masked gradient
10:  normalize factor scales and absorb them into the core
11:  increase  $P_{\text{act}}$  according to the continuation schedule, initializing the new coefficient at zero
12: end for
13: return  $\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta$ 

```

430 B Additional Proofs

431 B.1 Polynomial and Analytic Approximation

432 *Proof of Theorem 2.* Since g has degree at most P , write

$$g(s) = \sum_{p=0}^P c_p s^p. \quad (21)$$

433 Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^*$, and set $\beta_p = c_p$. Then, entrywise,

$$\hat{\mathcal{Y}}_{i_1, \dots, i_N} = \sum_{p=0}^P \beta_p (\mathcal{S}_{i_1, \dots, i_N}^*)^p = g(\mathcal{S}_{i_1, \dots, i_N}^*) = \mathcal{Y}_{i_1, \dots, i_N}. \quad (22)$$

434 Thus $\hat{\mathcal{Y}} = \mathcal{Y}$. $\square$

435 **Lemma 1** (Scalar-to-tensor approximation). *Let $\|\mathcal{S}^*\|_\infty \leq B$. For any scalar function g and any*
 436 *degree- P polynomial q_P ,*

$$\|g(\mathcal{S}^*) - q_P(\mathcal{S}^*)\|_F \leq \sqrt{\prod_{n=1}^N I_n} \sup_{|s| \leq B} |g(s) - q_P(s)|. \quad (23)$$

437 *Proof.* Every entry of $\mathcal{S}^*$ lies in $[-B, B]$, so every entrywise approximation error is bounded by
 438 $\sup_{|s| \leq B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the
 439 claim. $\square$

440 *Proof of Theorem 3.* Let $g(s) = \sum_{k=0}^\infty a_k s^k$ be the Taylor expansion of g around zero, and choose
 441 $q_P(s) = \sum_{k=0}^P a_k s^k$. By Cauchy's estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \quad (24)$$

442 For $|s| \leq B < \rho$,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^\infty |a_k| |s|^k \quad (25)$$

$$\leq M_\rho \sum_{k=P+1}^\infty (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \quad (26)$$

443 Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^*$, $\beta_k = a_k$ for $0 \leq k \leq P$, yields the stated unnormalized
 444 bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^N I_n$, gives the normalized form used in Theorem 3. $\square$

445 B.2 Rank Expansion

446 *Proof of Proposition 1.* Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1, \dots, i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}. \quad (27)$$

447 Taking an element-wise p -th power gives

$$(\mathcal{S}_{i_1, \dots, i_N})^p = \prod_{q=1}^p \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \prod_{n=1}^N A_{i_n, r_n^{(q)}}^{(n)} \right) \quad (28)$$

$$= \sum_{\gamma_1 \in [R_1]^p} \cdots \sum_{\gamma_N \in [R_N]^p} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \left(\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} \right), \quad (29)$$

448 where $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$. For each mode n , let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n ,
 449 so $|\alpha_n| = p$. Then

$$\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} = \prod_{r=1}^{R_n} (A_{i_n, r}^{(n)})^{\alpha_{n,r}} = A_{i_n, \alpha_n}^{(n)\langle p \rangle}. \quad (30)$$

450 Group all ordered tuples $(\gamma_1, \dots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \dots, \alpha_N)$, and define
 451 the aggregated core

$$\mathcal{X}_{\alpha_1, \dots, \alpha_N}^{(p)} = \sum_{\substack{\gamma_1, \dots, \gamma_N: \\ \text{count}(\gamma_n) = \alpha_n \ \forall n}} \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}}^{(q)}. \quad (31)$$

452 The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = \llbracket \mathcal{X}^{(p)}; \mathbf{A}^{(1)(p)}, \dots, \mathbf{A}^{(N)(p)} \rrbracket. \quad (32)$$

453 The n -th Veronese factor has $\binom{R_n+p-1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

454 B.3 Separation

455 **Lemma 2** (Full-rank Veronese evaluations). *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$
 456 whose Veronese feature matrix $\mathbf{A}^{(p)} \in \mathbb{R}^{D \times D}$ has rank D . Consequently, for $I \geq D$, a generic
 457 matrix $\mathbf{A} \in \mathbb{R}^{I \times R}$ has $\text{rank}(\mathbf{A}^{(p)}) = D$.*

458 *Proof.* The homogeneous degree- p monomials in R variables are linearly independent polynomials.
 459 Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^\alpha)_{i,\alpha}$, $|\alpha| = p$, is
 460 nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a
 461 nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the
 462 first claim.

463 For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix
 464 is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is
 465 therefore a generic property, so $\mathbf{A}^{(p)}$ has full column rank for generic $\mathbf{A}$. $\square$

466 *Proof of Theorem 1.* Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in$
 467 $\mathbb{R}^{I_n \times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^R a_r^{(1)} \otimes \dots \otimes a_r^{(N)}. \quad (33)$$

468 Its mode- n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)} \right)^\top. \quad (34)$$

469 For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R , and the Khatri-Rao product on the right has rank R . Thus
 470 $\text{rank}(\mathcal{S}_{(n)}) = R$ for every n , so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

471 Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree- p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1, \dots, i_N} = \left(\sum_{r=1}^R \prod_{n=1}^N A_{i_n, r}^{(n)} \right)^p \quad (35)$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^N \prod_{r=1}^R \left(A_{i_n, r}^{(n)} \right)^{\alpha_r}, \quad (36)$$

472 where $c_\alpha = p! / (\alpha_1! \dots \alpha_R!)$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)(p)} \in \mathbb{R}^{I_n \times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full
 473 column rank D for generic $\mathbf{A}^{(n)}$. The mode- n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \text{diag}(c) \left(\mathbf{B}^{(N)} \odot \dots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \dots \odot \mathbf{B}^{(1)} \right)^\top. \quad (37)$$

474 The diagonal matrix $\text{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices
 475 is full column rank, the right factor has rank D . Therefore $\text{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every
 476 mode n , and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

477 **B.4 Masked Generalization**

478 **Theorem 4** (Masked-entry uniform convergence). *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set*
 479 *with*

$$d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P + 1).$$

480 For $\theta \in \Theta$, define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \quad M = \prod_{n=1}^N I_n,$$

481 and the masked empirical risk

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^m \ell_{\mathbf{i}_t}(\theta),$$

482 where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \leq \ell_{\mathbf{i}}(\theta) \leq B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G -Lipschitz
 483 in θ , uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ , then with probability at
 484 least $1 - \delta$,

$$\sup_{\theta \in \Theta} |\mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta)| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

485 for every $0 < \varepsilon \leq B_\Theta$.

486 *Proof of Theorem 4.* Fix $0 < \varepsilon \leq B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε -net of Θ in Euclidean norm. Since Θ lies
 487 in a d -dimensional ball of radius B_Θ , it admits a net with

$$|\mathcal{N}_\varepsilon| \leq \left(\frac{3B_\Theta}{\varepsilon} \right)^d. \quad (38)$$

488 For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{\mathbf{i}_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$.
 489 Hoeffding's inequality gives

$$\Pr \left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \geq u \right) \leq 2 \exp \left(-\frac{2mu^2}{B_\ell^2} \right). \quad (39)$$

490 Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1 - \delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \leq B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \quad (40)$$

491 Substituting the net-size bound yields the stated stochastic term.

492 Now take any $\theta \in \Theta$, and choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \leq \varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \leq G\varepsilon, \quad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \leq G\varepsilon. \quad (41)$$

493 Therefore

$$\left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \quad (42)$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \quad (43)$$

$$\leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \quad (44)$$

494 Taking the supremum over $\theta \in \Theta$ completes the proof. $\square$

495 **Remark 1.** *The Lipschitz and bounded-loss assumptions are standard for compact parametric*
 496 *classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial*
 497 *coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may*
 498 *depend polynomially on these bounds and on P , but the covering dimension remains the number*
 499 *of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker*
 500 *expansion.*

C Additional Experiments

C.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 4 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^*)$. Let S^* be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^* = g(S^*) - \frac{\langle g(S^*), S^* \rangle}{\|S^*\|_F^2} S^*, \quad (45)$$

and, when $\|R^*\|_F > 0$, normalize it as

$$\tilde{R}^* = \frac{\|S^*\|_F}{\|R^*\|_F} R^*. \quad (46)$$

The observed tensor is then

$$Y_\alpha = \sqrt{1-\alpha} S^* + \sqrt{\alpha} \tilde{R}^*, \quad 0 \leq \alpha \leq 1. \quad (47)$$

By construction, $\langle S^*, \tilde{R}^* \rangle = 0$ and $\|\tilde{R}^*\|_F = \|S^*\|_F$, so α is exactly the energy fraction assigned to the nonlinear residual. This keeps the total energy fixed while moving the observation away from the linear Tucker backbone. The candidate residual families are $g(s) = s^2 + s^3$, $g(s) = (\exp(\kappa s) - 1)/\kappa$, and $g(s) = \tanh(\kappa s)$, with κ set from the input range.

The main text reports the nonlinear alpha sweep and degree sweep. Figure 6 shows that degree continuation activates higher powers without destabilizing the fit. Table 5 complements those figures with a small but consistent term-mass concentration effect: the dominant contribution usually lands on the degree that matches the injected response, while smooth odd links spread mass across nearby odd orders.

Table 4: Linear consistency on synthetic Tucker tensors. The $p > 1$ columns report effective higher-order NTD-PL contribution.

Method	bias = 0		bias = 0.5	
	RMSE↓	$p > 1$	RMSE↓	$p > 1$
Tucker	0.0114	–	0.0647	–
NTD-PL, $\beta_0 = 0$	0.0116	4.19%	0.0522	39.17%
NTD-PL, β_0 free	0.0117	3.19%	0.0118	2.98%

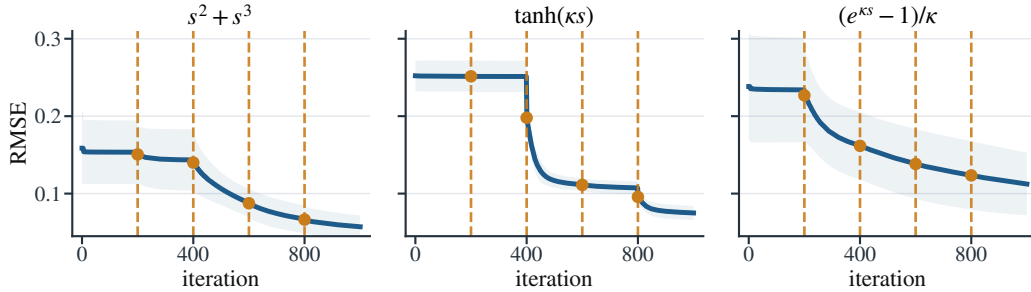


Figure 6: NTD-PL training curves for the controlled nonlinear mechanism checks at $\alpha = 0.25$ and $P = 5$. Dashed vertical lines mark degree-activation events.

Table 5: Representative effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at the matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p = 0$	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
s^2	0.5	43.3	23.2	5.8	8.2	9.1	9.8
$s^2 + s^3$	0.1	24.1	9.8	38.1	7.3	11.5	9.1
$\sin(\kappa s)$	0.1	37.7	0.6	35.4	2.8	19.8	3.6
$\tanh(\kappa s)$	0.2	37.9	1.3	30.9	5.4	17.3	7.0

521 C.2 CAVE Completion Significance and Robustness

522 The main text reports the full-resolution random-missing CAVE completion table at rank (33, 33, 4).
 523 We do not repeat that table here; instead, Table 6 gives the paired sign and Wilcoxon tests for the
 524 same protocol.

Table 6: Significance tests for full-resolution CAVE completion at rank (33, 33, 4). Metrics are computed on missing entries.

ρ	RMSE				SAM			
	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

Table 7: Link Refresh diagnostic on CAVE completion. Low/mid/high gains report the mean NTD-PL RMSE* gain over Tucker within each D_{link} tercile.

ρ	Spearman ρ_s	Low-yield RMSE* gain	Mid-yield RMSE* gain	High-yield RMSE* gain	Scenes
0.1	0.829	0.75%	7.92%	12.63%	15
0.3	0.871	-1.13%	9.10%	13.12%	15
0.5	0.889	-1.09%	9.16%	13.18%	15
0.7	0.868	-0.62%	9.22%	13.07%	15

525 C.3 Link Refresh Residual Diagnostic

526 The diagnostic used in the main text reuses the Link Refresh step from Section 5 as a one-round
 527 residual probe. Let $\hat{\mathcal{Y}}_{\text{T}}$ denote the Tucker output and $\mathcal{Y}$ the target tensor. Starting from this fitted
 528 Tucker latent tensor, we update only the scalar polynomial coefficients by fitting

$$h_{\gamma}(x) = \sum_{p=0}^4 \gamma_p x^p$$

529 by ridge regression on pairs $(\hat{Y}_{\text{T},i}, Y_i)$. In full reconstruction, all entries are used for this fit. In
 530 completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted
 531 map is applied entrywise to $\hat{\mathcal{Y}}_{\text{T}}$. We summarize this cheap refresh by the Link Refresh diagnostic

$$D_{\text{link}} = 10 \log_{10} \left(\frac{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{T}}\|_F^2}{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{link}}\|_F^2} \right),$$

532 which measures, on a log scale, how much Tucker residual energy is removed on the task-specific
 533 evaluation entries. In other words, it measures how much shared scalar-link yield is already present
 534 before the full NTD-PL alternating solver starts reshaping the Tucker parameters.

C.4 Joint Link-Family Smoke Test

The main model uses a polynomial link because it gives a closed-form ridge coefficient update and a direct Veronese rank-lift interpretation. To check whether the improvement is tied to the polynomial basis itself, we ran a small joint link-family smoke test on three CAVE scenes at spatial resolution 256×256 and backbone rank $(18, 18, 3)$. The spline variant replaces the polynomial link by a one-dimensional piecewise-linear link with eight knots. Its knot values are updated by ridge regression on the current latent entries, while the Tucker backbone is updated jointly through the link derivative.

Table 8: Joint scalar-link family smoke test on CAVE reconstruction. The spline link is slightly more flexible but slower; NTD-PL is the analyzable polynomial member used in the main paper.

Method	Params	RMSE↓	SAM↓	Time↓
Tucker	10k	0.0542 ± 0.0401	24.47 ± 10.95	$1.4 \pm 0.1s$
NTD-PL(P=4)	10k	0.0502 ± 0.0363	23.23 ± 10.85	$10.7 \pm 0.1s$
Joint Spline Link(K=8)	10k	0.0497 ± 0.0362	20.25 ± 8.01	$40.4 \pm 0.4s$

C.5 CAVE Beads Pseudo-RGB Visualization

Figure 7 visualizes the CAVE beads scene under the reconstruction and random-completion protocols. Each 31-band tensor is rendered as pseudo-RGB by selecting three representative spectral bands near 75%, 50%, and 20% of the wavelength index range and applying a shared linear intensity scaling within each task. For completion, unobserved entries in the observation panel are shown in gray.

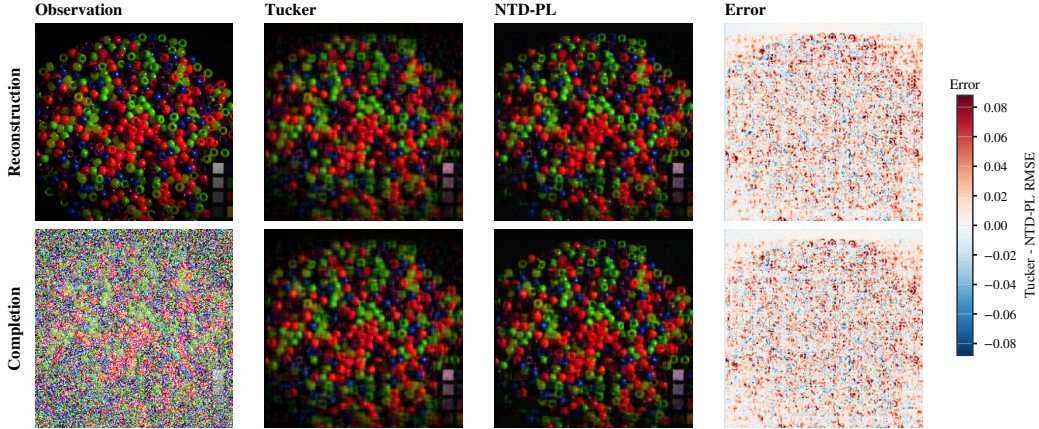


Figure 7: CAVE beads visual comparison. Rows correspond to reconstruction and random completion at missing rate 0.5. Error maps show Tucker minus NTD-PL per-pixel RMSE, so positive values indicate lower NTD-PL error.

C.6 Cross-Domain Real Tensor Checks

Table 9 reports matched-rank reconstruction on tensors outside the main hyperspectral setting: additional hyperspectral scenes, natural-image collections, and object-view tensors. The results are consistent with the scoped claim in Section 6.3: gains are clearest on hyperspectral and object-view data, while natural-image collections provide more modest improvements.

C.7 Non-HSI Link Refresh Diagnostics

The main diagnostic table uses per-unit non-HSI experiments rather than the collection-level tensors in Table 9. CBSD68 uses per-image units resized to $96 \times 96 \times 3$ with rank $(32, 32, 3)$; the available diagnostic CSV contains 16 CBSD68 units. CIFAR-10 uses 1000 class-balanced test images of shape $32 \times 32 \times 3$ with rank $(20, 20, 3)$. COIL-100 uses one object at a time as a $36 \times 48 \times 48 \times 3$ view tensor with rank $(8, 12, 12, 2)$. smallNORB uses one object instance at a time as an $18 \times 64 \times 64 \times 2$ stereo

Table 9: Additional matched-rank real tensor reconstruction. Columns report RMSE; gains are relative RMSE reductions over Tucker, and positive Δ RMSE favors NTD-PL.

Domain	Dataset	Shape	Rank	RMSE↓			Gain
				Tucker	NTD-PL	Δ	
Hyperspectral	Jasper Ridge	$100 \times 100 \times 198$	(11,11,43)	0.0462	0.0430	0.0032	6.9%
Hyperspectral	Samson	$95 \times 95 \times 156$	(10,10,35)	0.0263	0.0242	0.0021	7.9%
Hyperspectral	Urban	$307 \times 307 \times 162$	(43,43,47)	0.0444	0.0428	0.0016	3.6%
Natural images	CBSD68	$68 \times 96 \times 96 \times 3$	(20,32,32,3)	0.1327	0.1307	0.0020	1.5%
Natural images	CIFAR-10	$1000 \times 32 \times 32 \times 3$	(96,20,20,3)	0.0753	0.0724	0.0029	3.8%
Object views	COIL-100	$8 \times 36 \times 48 \times 48 \times 3$	(4,8,12,12,2)	0.0991	0.0931	0.0060	6.0%
Object views	smallNORB	$25 \times 18 \times 64 \times 64 \times 2$	(18,12,24,24,2)	0.0454	0.0429	0.0024	5.4%

view tensor with rank (12, 24, 24, 2). All units are max-normalized independently, and diagnostic labels are assigned only from D_{link} , not from NTD-PL gain.

Table 10: Representative non-HSI units selected by D_{link} . Boundary/effective labels are assigned before inspecting NTD-PL gain.

Domain	Dataset	Unit	Label	D_{link}	Tucker RMSE	NTD-PL RMSE	Gain
Natural images	CIFAR-10	cifar10_test_06970	boundary	-0.136	0.0008	0.0008	-4.46%
Natural images	CBSD68	cbsd68_011	moderate	0.077	0.0409	0.0392	4.25%
Natural images	CBSD68	cbsd68_012	effective	0.341	0.0490	0.0408	16.69%
Object-view	COIL-100	coil_obj088	boundary	0.017	0.0280	0.0276	1.51%
Object-view	COIL-100	coil_obj091	effective	0.875	0.0751	0.0650	13.34%
Object-view	smallNORB	smallnorb_c3_i4	boundary	0.077	0.0315	0.0301	4.51%
Object-view	smallNORB	smallnorb_c2_i7	effective	1.048	0.0478	0.0380	20.53%

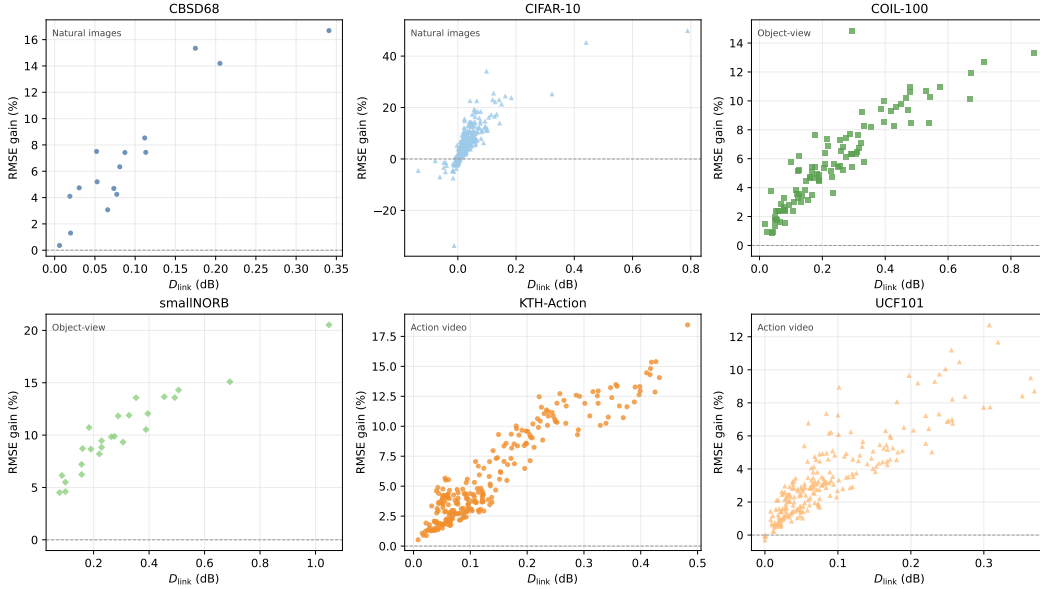


Figure 8: D_{link} versus NTD-PL RMSE gain over non-HSI tensor units, shown as per-dataset panels for natural-image, object-view, and action-video settings. Units are selected independently and labels are assigned from D_{link} .

561 NeurIPS Paper Checklist

562 This draft uses a working checklist. It should be replaced by the official NeurIPS checklist distributed
563 with the final 2026 submission template.

- 564 1. **Claims.** The main claims are stated in the abstract, introduction, theory section, and experiment
565 captions: NTD-PL is a Tucker enhancement and shared-backbone rank lift for Tucker tensors; its
566 gains are tied to jointly reshaping the Tucker backbone under a shared scalar link rather than to
567 modest rank increases alone.
- 568 2. **Limitations.** Section 7 discusses predicted near-linear boundary cases, whole-axis missingness,
569 shared-link assumptions, and the current domain concentration in hyperspectral data.
- 570 3. **Theory assumptions.** The approximation results assume a bounded latent tensor and polynomial
571 or analytic scalar links. The separation results use generic full-rank feature constructions. The
572 masked generalization bound assumes bounded parameters, bounded observations, and random
573 observed entries.
- 574 4. **Reproducibility.** The experiment section reports datasets, ranks, mask types, missing rates,
575 and baseline definitions. The codebase includes scripts for rank sweeps, MLPCal, structured
576 missingness, and the cross-domain real-tensor reconstruction study; the final version should add
577 hardware details and exact command manifests.